

1. (50 points) The reaction rate of a 1st-order, elementary catalytic reaction ($A \rightarrow B$) was measured in a well-stirred reactor in which 5 grams of 0.1 mm catalyst particles were suspended. For a concentration of reactant of 1×10^{-3} mol/L, the reaction rate was found to be 1×10^{-4} mol/g/s.

- When the stirring rate was increased, the reaction rate did not change. What does it say about the reaction?
- Show that these data were obtained without the effects of pore-diffusion limitation.
- This reaction is to be run in a packed-bed reactor that is 10 m long and contains 10 cm diameter tubes. What particle diameter should be used in this reactor to get 75% conversion? Will there be pore-diffusion limitation? Assume that there is no pressure drop.

Useful information:

$D_e = 10^{-1}$ cm²/s (De is constant no matter what size particles are used)

ρ_B = (bulk density of catalyst in the packed bed) = 1 g/cm³ (same for any particle size)

ρ_C (catalyst density) = 2 g/cm³ (same for any particle size)

$C_{A0} = 1 \times 10^{-3}$ mol/L

$F_{A0} = 1$ mol/s

a) The rate of mass transfer is so high that the rate of stirring is negligible. The reaction happens very quick.

b) Using Weisz - Prater Criterion

$$C_{wp} = \frac{-r_A \rho_C R^2}{D_e C_{As}} = \frac{(1 \times 10^{-4} \frac{\text{mol}}{\text{g} \cdot \text{s}}) (2 \frac{\text{g}}{\text{cm}^3}) (0.005 \text{ cm})^2}{10^{-1} \frac{\text{cm}^2}{\text{s}} \cdot (1 \times 10^{-3} \frac{\text{mol}}{\text{L}}) (\frac{1 \text{ L}}{1000 \text{ cm}^3})} = 0.05$$

$C_{wp} \ll 1 \Rightarrow$ no internal pore diffusion limitation.

c) $F_{A0} \frac{dx}{dW} = -r'_A \eta = -k' C_{A0} (1-x) \eta$

- solving for effectiveness factor

$$\eta = \frac{F_{A0}}{W k' C_{A0}} (-\ln(1-x)) \quad (1)$$

+ Finding W

$$W = V \times \rho_B = \left(10 \text{ m} \times \frac{100 \text{ cm}}{\text{m}}\right) (5 \text{ cm})^2 (\pi) (1 \text{ g/cm}^3)$$

$$W = 78539.82 \text{ g} \quad (2)$$

+ Finding k'

$$k' = 10^{-4} \frac{\text{mol}}{\text{g} \cdot \text{s}} \times \left(\frac{\text{L}}{10^{-3} \text{ mol}}\right) = 0.1 \frac{\text{L}}{\text{g} \cdot \text{s}} \quad (3)$$

Plug (2) & (3) into (1)

$$\eta = \frac{F_{A0}}{W k' C_{A0}} (-\ln(1-x)) = \frac{(1 \text{ mol/s})}{(78539.82 \text{ g}) (0.1 \text{ L/g} \cdot \text{s}) (1 \times 10^{-3} \frac{\text{mol}}{\text{L}})} (-\ln(1-0.75))$$

$$\eta = 0.17651$$

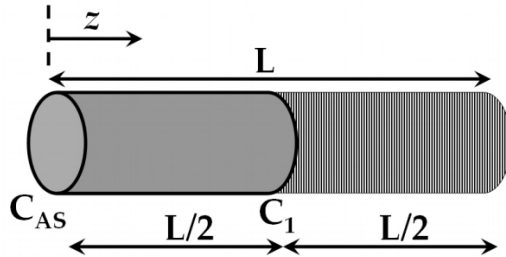
$$\phi = \frac{1}{0.17651} = 5.665$$

$$\phi = \frac{R}{3} \sqrt{\frac{k' \rho_C}{D_e}} \Rightarrow R = \frac{3 \phi}{\sqrt{\frac{k' \rho_C}{D_e}}} = \frac{3(5.665)}{\sqrt{\frac{(0.1 \frac{\text{L}}{\text{g} \cdot \text{s}}) (1000 \frac{\text{cm}^3}{\text{L}}) \cdot 2 \text{ g}}{10^{-1} \text{ cm}^2/\text{s}}}} \Rightarrow R = 0.38 \text{ cm}$$

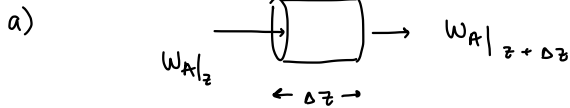
$$C_{wp} = \eta \times \phi_1^2 = \frac{-r_A' \int_0^R R^2}{D_e C_{AS}} = 0.17651 \times 5.665^2$$

$$C_{wp} = 5.645 > 1 \Rightarrow \text{internal diffusion limits the reaction severely}$$

2. (50 points) A pore of diameter d and length L on catalyst surface has degraded over time and is now half poisoned. Due to poisoning, the first half of the pore (till $z = L/2$) is non-reactive. The concentration at the inlet is C_{AS} and at the $L/2$ is C_1 (known). The diffusivity of the reactant is D_e . If the reaction is a first order reaction on the surface with a rate constant $= k''$ (length/time units) then:



- Solve for concentration profile from $z = 0$ to $L/2$. State all the assumptions.
- If the effectiveness factor for no poisoning is $\eta_1 = \frac{\tanh(\phi_1)}{\phi_1}$, where $\phi_1 = L \sqrt{\frac{4k''}{dD_e}}$; calculate the new effectiveness factor for poisoning in terms of C_{AS} , C_1 and ϕ_1 .



no reaction : $W_A \left(\frac{\pi d^2}{4} \right) \Big|_z - W_A \left(\frac{\pi d^2}{4} \right) \Big|_{z+\Delta z} + \cancel{r_A' \left(\frac{\pi d^2 \Delta z}{4} \right)} = 0$

no reaction $\rightarrow$ no accumulation

$$W_A \Big|_z - W_A \Big|_{z+\Delta z} = 0 \quad (\text{mass transfer at inlet and outlet is the same})$$

$$\rightarrow \int \frac{dW_A}{dz} = 0 \rightarrow W_A = k_1$$

$$W_A = -m \frac{dC}{dz} \rightarrow k_1 = \frac{dC}{dz}$$

$$\rightarrow C = k_1' z + k_2$$

BC 1 : $C \Big|_{z=0} = C_{AS}$
 $\Rightarrow C_{AS} = k_2$

BC 2 : $C \Big|_{z=\frac{L}{2}} = C_1$
 $\rightarrow C_1 = k_1' \frac{L}{2} + C_{AS}$
 $\rightarrow k_1' = \frac{C_1 - C_{AS}}{L/2}$

$$C = \frac{C_1 - C_{AS}}{L/2} z + C_{AS}$$

$$b) \quad \phi_{\text{non-poisoned}} = \left(\frac{L}{2} \right) \sqrt{\frac{4k''}{aDe}} = \frac{\phi_1}{2}$$

$$\eta_{\text{non-poisoned}} = \frac{\tanh\left(\frac{\phi_1}{2}\right)}{\left(\frac{\phi_1}{2}\right)}$$

overall η

$$\eta \left(\pi d L \right) \left(k'' C_{AS} \right) = \eta_{\text{non-poisoned}} \left(\pi d L / 2 \right) \left(k'' C_1 \right)$$

$$\eta = \eta_{\text{non-poisoned}} \cdot \frac{C_1}{2C_{AS}}$$

$$\boxed{\eta = \frac{\tanh\left(\phi_1/2\right)}{\phi_1} \cdot \frac{C_1}{C_{AS}}}$$